

NASA Astronomy Briefing — MVP Design Doc

What this is:

A deliberately small, source-grounded NASA news briefing and question app. The app supports questions about recently indexed NASA news and follow-up questions on returned stories. Note that it is not a search of NASA's complete historical archive.

Scope:

NASA news only for the MUP; full article bodies, not RSS snippets.

1. High-level: how does the app work?

A user asks for recent NASA news. The app returns a short, source-linked briefing. The user can then ask for more detail about an item, or ask whether the stored NASA news covers a topic they have heard about, such as the 3I/ATLAS comet.

- **User does:** asks for recent NASA / astronomy news.
- **App returns:** a short, grounded answer with links to the NASA article(s) used.
- **Rough flow:** (user → route → SQL or semantic retrieval or both → answer + citations).

2. The data

- **What is the data?** Full text of NASA news articles plus source and publication metadata.
- **Where does it come from (the source)?** NASA news RSS, followed by the linked NASA article page. I have ran a few brief test spikes using Node and OpenAI to pull the info, so it is viable. Later, I would loo to integrate the European Space Agency RSS feed. Also arXiv is a public library of astronomy papers we could potentially access via API (MUCH later).
- **How do you plan to get it?** Scheduled ingestion gets RSS items, fetches/extracts the full article, and stores it in SQL.

3. Freshness

- **Does the data need to be up to date?** It does! Daily freshness for MUP, to cover any changes.
- **Initial coverage:** Before daily ingestion begins, I would backfill the most recent 90 days of NASA news (or more, depending on testing general queries). The app will state its indexed coverage period rather than implying it contains all historical NASA material. Then, we can have a user ask questions about slightly older things and receive grounded answers.
- **Non-recent topics:** The app will search only the NASA news currently indexed. If a topic predates the coverage window and no relevant article is found, it will return a clear no-result message rather than using unsourced general knowledge, or implying NASA has never covered it.
- **Enforcing idempotency :** for SQL, we compare the URL and stored content hash. For Pinecone, we use vector IDs. A daily job upserts by canonical URL and content hash; failures logged and retried without overwriting good content.

4. Chunking & metadata

- **Chunking strategy:** split cleaned full article text into (slightly) overlapping, paragraph-aware chunks; keep title context with each chunk.
- **Metadata stored per chunk:** article ID, chunk index, source, published date, and content hash.

5. Vector store

- **Which vector store, and why:** Pinecone for managed semantic similarity search, as it's what I am experienced with. SQL remains canonical for exact dates, filters, and content.
- **Embedding model:** model: 'text-embedding-3-small', dimensions: 1536.

6. Agent architecture / workflow

- **Is there an agent, or just retrieve → answer?** No autonomous agent, just retrieve → answer. Rules route date/recent questions to SQL and topic questions to Pinecone. Same

kind of selector – RAG / SQL / hybrid – Aggregator-type setup as before.

- **Tools the workflow uses:** SQL date/source queries and Pinecone semantic retrieval; no open-web tool or autonomous agent (MUP).
- **Semantic retrieval and reranking:** For a topic question, Pinecone retrieves a small set of candidate chunks. A reranking step can reorder those candidates before the app builds its final answer, using the user's exact question and the retrieved text. For the MUP, start with Pinecone's top results; add reranking only if evaluation shows relevant chunks are not consistently appearing near the top.
- **Outside-corpus questions:** If no relevant stored NASA result is found, the app says: "I couldn't find this in the NASA news currently indexed. This currently only answers from its stored NASA news sources." A general LLM answer is deferred because it might be stale. And we cannot chunk ALL the NASA corpus of news related to everything.

7. Observability & evals

- **How will you know it's working?** A small set of questions – expected source checks, duplicate-ingestion and idempotency tests, logs for ingestion/retrieval IDs and errors.
 - **Deferred for a later version:** ingesting ESA feeds, arXiv papers, checking the live Launch Library, scraping YouTube transcripts of astronomy creators I enjoy, adding authentication, live launch information, maybe an opt-in daily/weekly email briefing (*kind of human in the loop-ish*) – but all this only after this NASA slice works reliably.
 - **Later feedback loop:** Maybe let users/subscribers rate a briefing or mark stories as useful/not useful. Use that feedback to evaluate briefing quality; do not automatically retrain or change retrieval behaviour from individual feedback.
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